

Course Project

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Bench # 9

Circuit Theory I

Section 803

Course number (EECE.2070)

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Honor Pledge: I promise that I will not give or receive any unauthorized help in performing any portion of this project and that all work will be my own.

I: Summary

A simple three-level water level indication system using a 9 V battery as the source has been implemented. Three BC547 NPN transistors are acting as low-side switches for three LEDs, each connected to a 330-ohm resistor. Hence, an immediate indication can be viewed on the display as and when the water level escalates. It complies with course requirements regarding working with a 9 V source and developing an observable result, such as LEDs, in a creative project. The design depicts an electronic system that uses sensors to detect the height of water in a tank and shows the current level via LEDs, a buzzer, or another form of output. It simulates how real-world automated tank monitoring systems operate by turning water levels into electrical signals that activate various indicators. It was designed to give a simple, dependable approach to detecting water levels without the need for frequent manual checks. The design prevents tanks from overflowing or going dry by providing timely and unambiguous feedback on the water status. Beyond the course requirement, the goal is to demonstrate a viable solution for improving water management in homes, commercial buildings, and industrial systems. It demonstrates how sensors, control logic, and output devices can be used to automate repetitive activities and eliminate human error. The purpose of the water-level indicator is to monitor the water height in a tank and alert users when it reaches specific thresholds—such as “low,” “mid,” or “full.” This helps maintain efficient water usage, avoid shortages, and prevent overflow or pump damage. This promotes effective water usage, avoids shortages, and protects against overflow or pump damage.

II: Equipment

Table 1: Equipment Used

Equipment Type	Details
Dual Power Supply	Make: GWINSTEK Model: GPD-3303D Serial #: EL832106
DMM Bench Top	Make: Keithley Model: 2110 5 ½ Serial #: 8007796
DMM Hand Held	Make: Tenma Model: 72-9385 SPC Type #: DB0E48
Analog Discovery Kit 2	Make: Digilent Model: ADK2 Serial #: DB0E48
• Bench Shoebox • Breadboard • Safety glasses • Soldering Kit	N/A

Table 2: Components Used

Components	Quantity
9-volt Battery	1
BC 547 Transistors	3
330 ohm resistors	3
LED	3
Assorted wires	As needed

III: Introduction

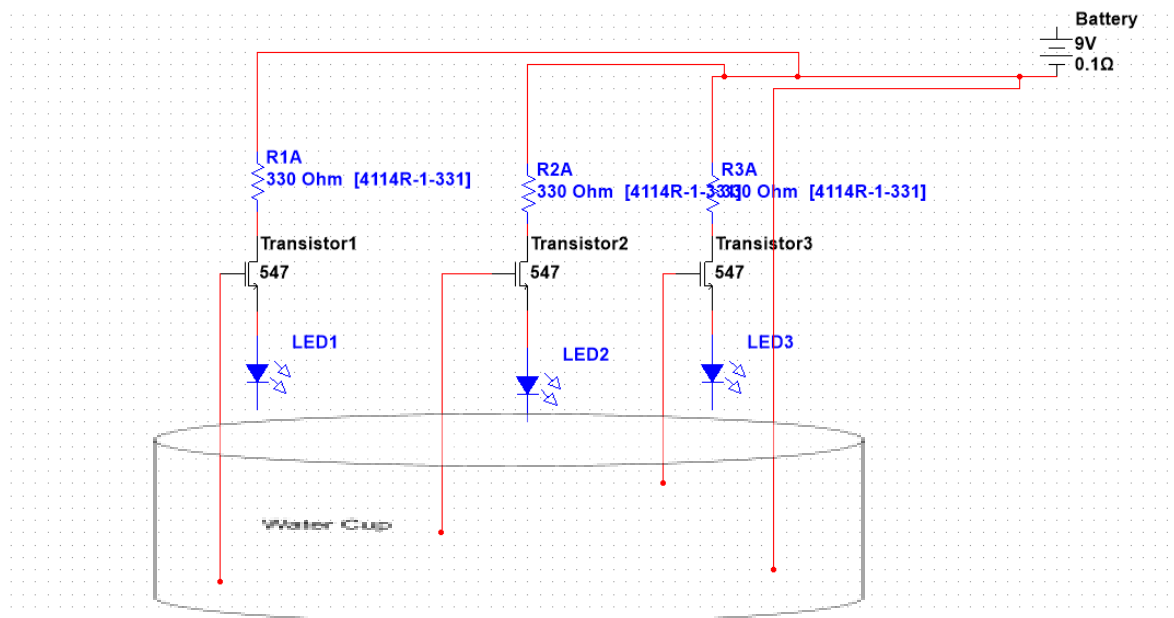
The specific goal of this lab was to design, build, and verify the operation of a three-level water level indicator that uses a 9 V supply and discrete components to show low, medium, and high water levels using LEDs. Practically, we were trying to determine whether water contact at different probe heights could reliably create a small sensing signal and whether that signal could be amplified through transistors to produce a clear, repeatable visual output.

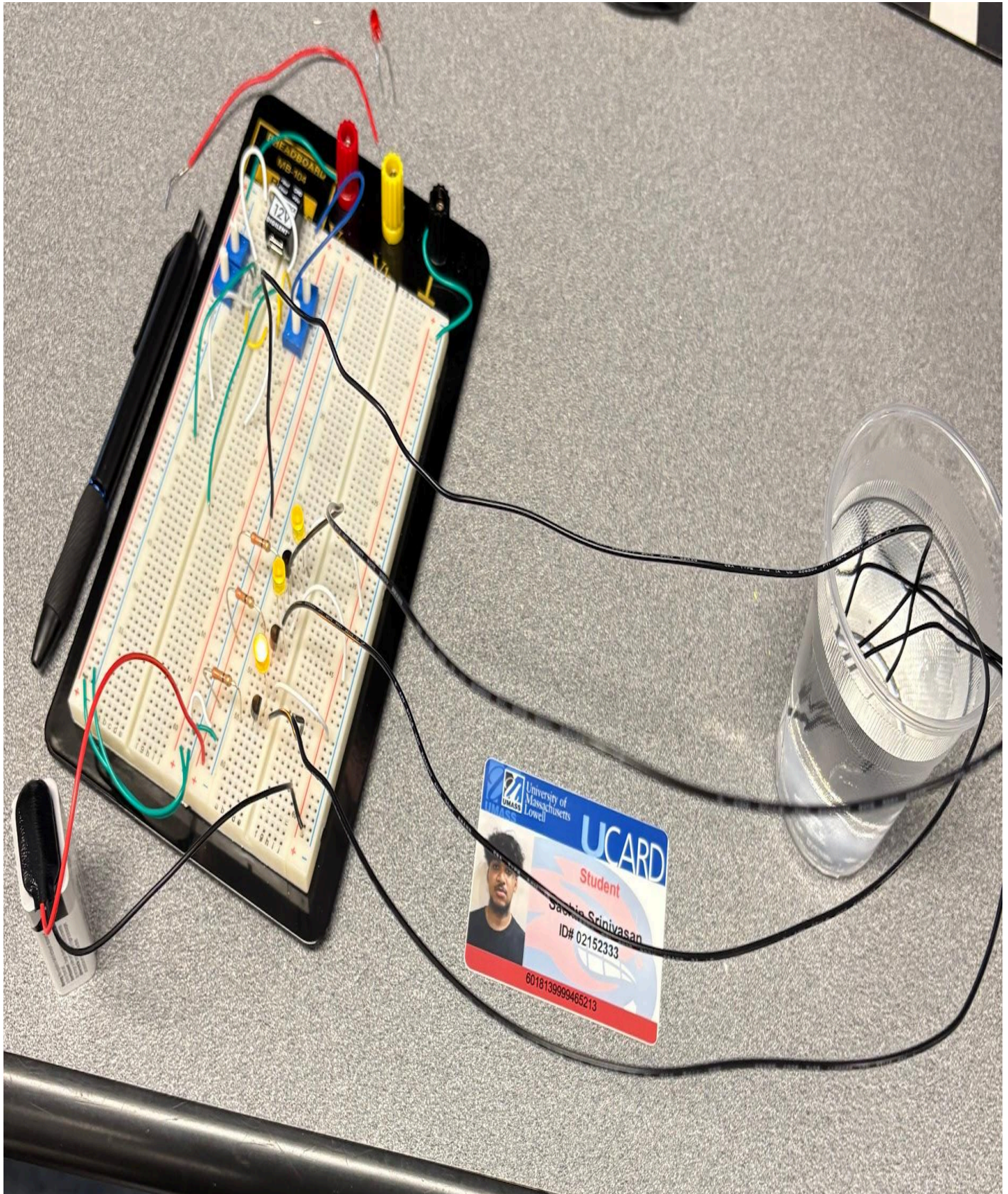
Scientific information already established for this topic includes (1) the idea that most real-world water contains ions and therefore has finite conductivity, allowing current to flow between submerged probes, and (2) the theory of BJT transistor operation, where a small base current controls a larger collector current. The key theories and relationships used are the BJT “switch” concept, basic circuit laws such as Ohm’s Law $V=IR$, and current limiting for LEDs using a resistor where the LED current is approximately $I = (V_{\text{Supply}} - V_{\text{LED}} - V_{\text{CE}}) / R$.

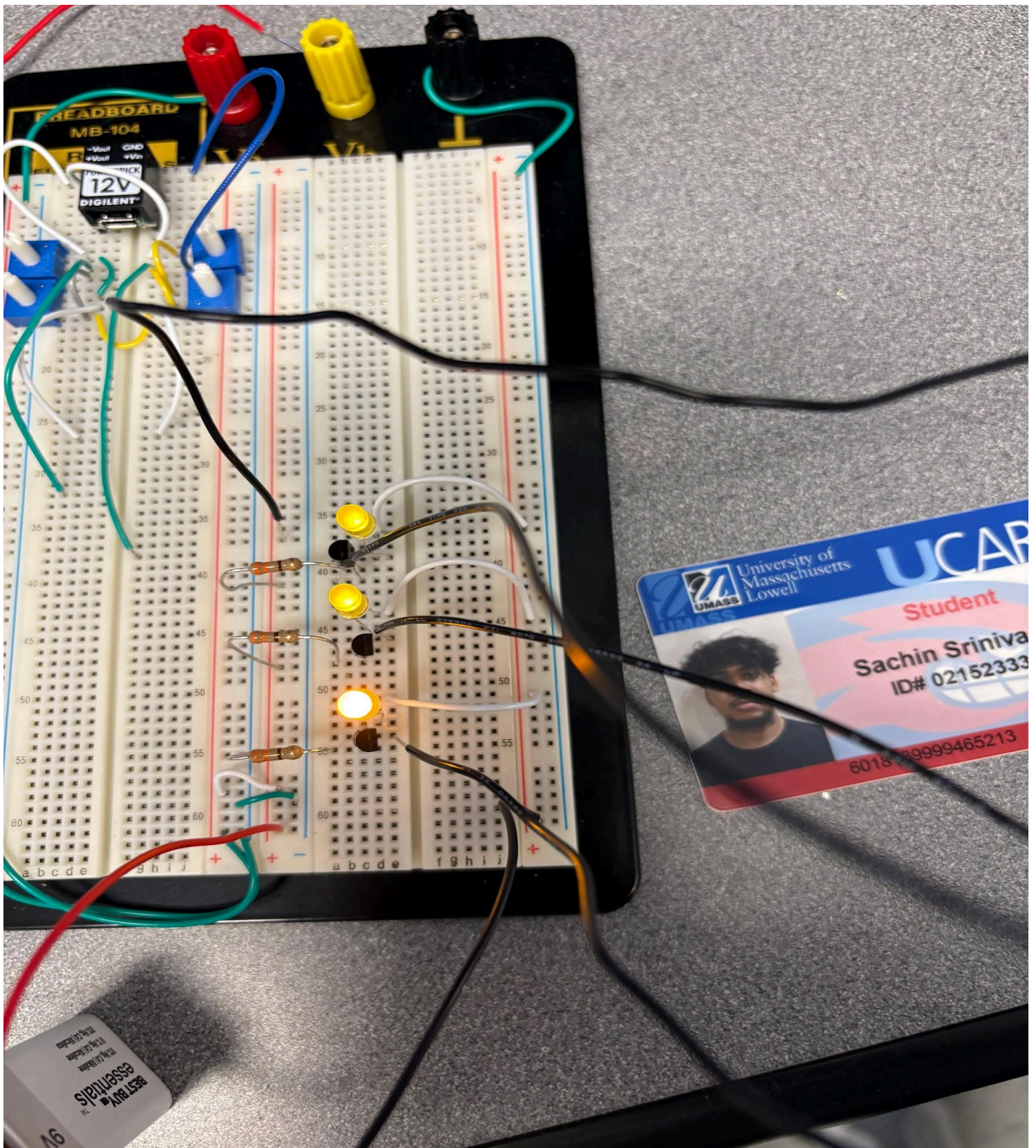
This lab goal connects directly to the theory because the experiment uses water's conductivity to provide the small base drive needed to turn each BC547 transistor on, demonstrating signal amplification and switching: a weak sensing path through water triggers a transistor to saturate and allow a much larger current through an LED-resistor branch. As the water level rises and reaches each successive probe, the circuit transitions from cutoff to saturation for that stage, illustrating how BJT switching and resistor-based current control can be used to convert a physical condition of the water height into an electrical output.

IV: Circuit Description

Expected Values: 9V to each LED. This was achieved when taking my measurements; the voltage measurement for each LED was around 9V.







V: Measurement

The completed circuit functioned as a three-stage water level indicator: as the water rose, the LEDs turned on in sequence when the water reached each probe height. Each LED consistently lit only after its corresponding probe made contact with the water, showing a clear low/medium/high level indication. The LED brightness remained generally steady during normal operation, confirming the transistors were switching properly and the $330\ \Omega$ resistors were limiting current.

VI: Discussion

The sensor consisted of components assembled on a breadboard with a 9 V battery, three LEDs, three 330-ohm resistors, and three BC547 transistors. The common reference probe was positioned near the bottom of the container and grounded with reference to the sensor ground or common reference, while three sensing probes were arranged at various heights, ranging from low, mid, and high. To a BC547 transistor, each sensing probe is connected, so that as water approaches a probe, it makes a resistive path from the probe to and common reference probe, completing a small current flow to the base. Due to the small current flow, the transistor becomes saturated or switched on as a switch. As a result, with the transistor on, current flows from +9 V through an LED and a 330-ohm resistor and then Collector and Emitter paths to ground, lighting an LED represents its respective level. However, normally, an LED current would be limited solely by a resistor. To test functionality, water was raised and ensured that respective LEDs lit up progressively as probes became submerged beneath them, as per specifications necessary for implementing a “do something” project requirement.

VI: Conclusion

A three-stage water level indicator with a functional water level display system was successfully implemented with fewer discrete components.

The display system shown illustrates the possibility of biasing a BJT with low sensor current flowing through water, thus displaying an unmistakable indication of low/medium/high water levels with an LED indication with a 9-V supply as required, and an unmistakable observable output. The project meets the stipulated importance of the working functionality on an electronic principle, like displaying LEDs, and forms a platform for improvements on a probe guarantee, corrosion prevention, and alert outputs.

VIII: Questions

Are all the answers to the experimental questions indicated throughout the experimental procedure in this section? Are they correct?

Yes, all the experimental questions are indicated throughout the experimental procedure.

IX: References

- 1) None